Perceived Authorship and Trust: A Pilot Study on Human vs AI-generated News Writing

Perceived authorship and trust

How Perception Shapes Trust More Than Actual Authorship

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Figure 1: Illustration of human evaluation of AI-generated and human-written content in a controlled study setting.

As artificial intelligence becomes increasingly integrated into journalism, questions around trust and credibility in AI-generated content continue to emerge. This study explores how perceived authorship—whether a reader believes a text is written by a human or an AI—affects trust in news writing. In a controlled pilot study (n = 10), participants read four short news articles about a university event, two written by humans and two generated by AI systems. Participants were given varying instructions about authorship and asked to evaluate each article's trust and credibility. Results show minimal differences in trust between AI and human-written articles based on actual authorship. However, participants frequently misidentified authorship and appeared to rely on subjective perceptions rather than accurate detection. These findings suggest that trust in news content may be influenced more by perceived authorship than by actual origin, raising important considerations for human-centered AI design in journalism.

CCS CONCEPTS
• Human-centered computing → Human computer interaction (HCI)
• Human-centered computing → User studies

Keywords: AI-generated text, trust, authorship perception, journalism, human-centered AI

1. INTRODUCTION

The increasing use of artificial intelligence in content creation has introduced new questions about trust, credibility and authorship in journalism. As AI systems become capable of producing fluent, structured and factually coherent news-style writing, readers are often unable to easily distinguish between human- and AI-generated content.

This raises an important question: does trust depend on who actually wrote the content, or on what readers believe about its authorship?

This study investigates how perceived authorship influences trust in news writing. Rather than focusing solely on whether participants can accurately identify AI-generated text, this research examines how trust shifts based on both actual and perceived authorship.

Through a small-scale experimental study, participants evaluated four news articles and reported their perceptions of trust, credibility and authorship. By comparing these responses, this study aims to better understand how readers interpret and evaluate AI-generated journalism in a human-centered context.

2. METHOD

2.1 Participants

A total of 10 participants took part in the study. Participants came from varied professional backgrounds and completed the study individually through a Google Form.

2.2 Materials

Participants were presented with four short news articles (~150–200 words each) about a Northeastern University workshop on the Retro Mobile Gaming Project.

- Two articles were written by human authors
- Two articles were generated using an AI system

All articles were written in a similar journalistic style to maintain consistency in tone, structure and length.

2.3 Procedure

Participants received instructions indicating whether each article was written by a human or an AI system. These labels varied across participants.

For each article, participants were asked to:

- Rate trust (scale of 1–5)
- Rate credibility (scale of 1–5)
- Recall what they were told about authorship
- Indicate who they believed wrote the article (AI or human)
- Optionally explain their reasoning

Each participant evaluated all four articles.

2.4 Results

Trust scores were averaged across articles to compare:

- Human vs. AI-written articles (actual authorship)
- Participant perceptions of authorship

Given the small sample size, analysis focused on identifying patterns rather than statistical significance.

3. RESULTS

3.1 Trust by actual authorship

Participants rated all four articles similarly in terms of trust.

- Human-written articles received an average trust score of **3.65**
- AI-generated articles received an average trust score of **3.6**

The difference between these values was minimal, indicating that participants did not significantly distinguish between AI and human writing based on trust.

3.2 Article-level trust

- Article 1 (Human): **3.9**
- Article 2 (Human): 3.4
- Article 3 (AI): **3.8**
- Article 4 (AI): 3.4

The highest-rated article was human-written (3.9), but one AI-generated article received a nearly identical score (3.8).

3.3 Perception of authorship

Participants showed inconsistency in identifying authorship. In multiple cases:

- AI-generated articles were perceived as human-written
- Human-written articles were perceived as AI-generated

These misclassifications suggest that participants relied on subjective cues such as tone, clarity and structure rather than accurately identifying authorship.

3.4 Perception vs instruction

Analysis of participant responses shows a divergence between assigned labels and perceived authorship.

For Article 1 (human-written), participants were evenly split in what they were told (50% AI, 50% human), yet 70% believed the article was written by a human.

For Article 3 (AI-generated), 80% of participants were told the article was AI-generated. However, 60% believed it was written by a human.

These results indicate that participants frequently relied on their own interpretation of the text rather than the provided label.

Notably, Article 3 (AI-generated) received a trust score of 3.8, nearly identical to the highest-rated human-written article (3.9), reinforcing the difficulty participants had in distinguishing between AI and human writing.

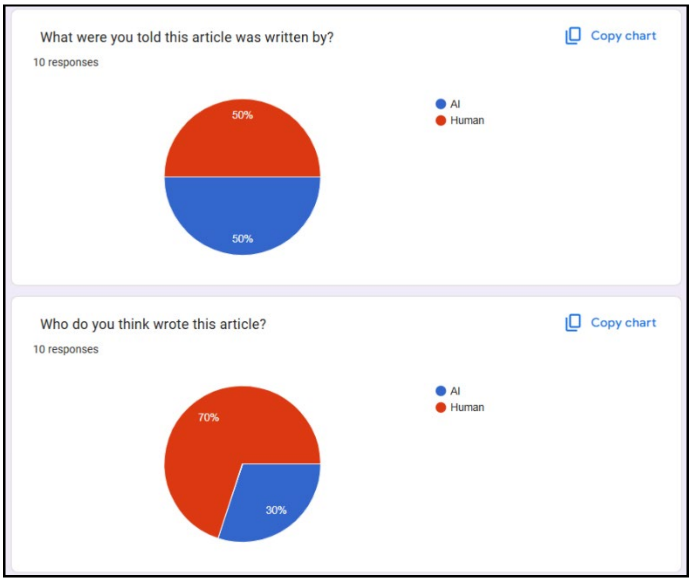

Figure 2: Perceived vs assigned authorship for Article 1 (human-written)

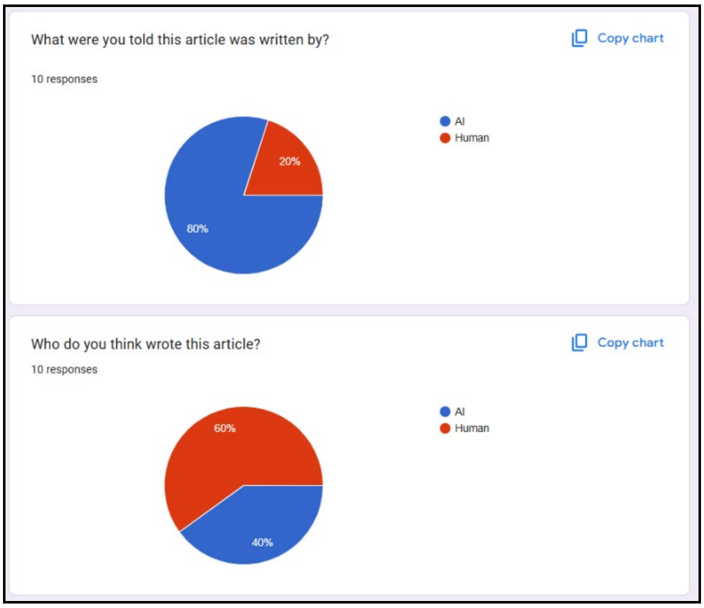

Figure 3: Perceived vs assigned authorship for Article 3 (AI-generated)

4. DISCUSSION

The results of this study highlight a key distinction between actual and perceived authorship in shaping trust. The findings suggest that trust in news writing is not strongly determined by whether content is written by a human or an AI system. Instead, participants appeared to evaluate articles based on perceived quality and subjective interpretation.

The similarity in trust scores across all four articles indicates that AI-generated writing can match human-written content in perceived credibility, at least in controlled settings.relied

Additionally, the frequent misidentification of authorship highlights a key challenge: readers may not be able to reliably distinguish between AI and human writing, even when prompted to do so.

This has important implications for human-centered AI design. If users rely on perception rather than accurate detection, then transparency, labeling and communication strategies become critical in shaping trust.

5. LIMITATIONS

This study has several limitations.

First, the sample size was small (n = 10), which limits generalizability. Second, all articles covered the same topic and were written in a similar style, which may have reduced variability in responses. Third, participants completed the study in a controlled format, which may not reflect real-world reading behavior.

Future work could expand this study with larger samples, more diverse content and varied writing styles.

6. CONCLUSION

This study explored how perceived authorship influences trust in news writing. Results show that participants did not significantly differentiate between AI-generated and human-written articles based on trust scores alone.

Instead, trust appeared to be shaped more by perception than by actual authorship. As AI continues to play a larger role in content creation, understanding how readers interpret and evaluate authorship will be essential for building trustworthy human-AI systems.

##References

None